

Assignment 2 – Student Performance Analysis

This report explains the full implementation of the Student Performance Analysis system.

1. Campus Data Files:

Three sample campus files (CMP001.txt, CMP002.txt, CMP003.txt) were created, each containing five students with marks in:

- Programming
- Database
- Systems Analysis
- Networking
- Student ID

2. Python Program Overview:

The Python solution performs:

- Automatic detection and loading of all campus files.
- Calculation of average marks for each subject in each campus.
- Identification of At-Risk students (average score < 50).
- Generation of a summary file per campus.
- Creation of a consolidated academic_performance_report.txt.
- Implementation of interactive keyword search.

3. Summary Files:

Each campus produces:

CAMPUS_CODE_summary.txt containing:

- Subject averages
- List of At-Risk students

4. Academic Performance Report:

The generated report includes:

- Total campuses processed

- Total students processed
- Overall subject averages
- High performer percentage
- At-risk percentage
- Recommendations

5. Keyword Search Feature:

The program loads the academic report and allows the user to enter any keyword, counting occurrences (case-insensitive). Typing 'exit' closes the search.

This PDF accompanies the completed Python code and sample input data files as required.